Grid Integration of Residential Solar

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Abstract

Rooftop solar and home battery systems change how the utility company balances the grid. We study net metering, inverter export limits and battery storage in 400 homes.

Results

Homes with a lithium-ion battery exported 40 percent less solar energy at midday.

Smart inverter settings reduced voltage problems on the grid.

Heat pump adoption shifted electricity consumption to winter evenings.

References

[1] Battery Storage Economics, internal memo.
[2] https://doi.org/10.5555/other.paper